

StoreFlow Analytics Delivery 1

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Introduction

StoreFlow Analytics

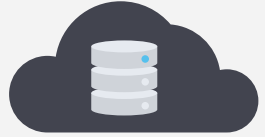
StoreFlow Analytics is a retail analytics system designed to help independent store owners understand customer behaviour inside their stores by combining traced camera input, aggregate customer profiles, and sales data.

Intended Users

- Small retail operators
- Independent grocery store owners
- Store managers looking to improve layout, staffing, and product placement

Major Functionality

- *File*
- *Help*
- *View*
 - Colour Themes
- *Camera*
 - Select and toggle active camera





Customer Interview



- Owner wants details on store's conversion rate
 - Visitors $\leftrightarrow$ Purchasers
- Owner wants to know if product placement or shelf advertising affects sales
- Owner wants additional store visibility on:
 - Customer movement and foot traffic
 - Areas of congestion
 - Cashier wait times
- Preference of customer privacy $\rightarrow$ no identifiable tracking
- Owner wants to integrate existing sales data infrastructure



Vision and Stories' Summary

Vision

The system helps independent retail stores understand customer behaviour, movement/traffic, and purchasing outcomes so they can improve layout, promotions, and operations using real data instead of guesswork

Stories

1. Store Layout / Mapping
 - a. A digital 2D layout of the store so customer movement can be mapped to real locations.
2. Customer Behaviour Tracking
 - a. Monitor how customers move through aisles to understand traffic patterns and behaviour.
3. Sales + Behaviour Data Integration
 - a. Combine movement data with cashier sales data to analyze conversion rates.
4. Analytics & Actionable Insights
 - a. Generate reports on wait times, congestion, and product interaction to guide operational decisions.



Development/Demo – Story Assignment

Compiled Binaries for Windows x86

P: High

Cost: 1.5 Days



Wiki/ReadMe

P: Medium

Cost: 1.5 Days



Mock Sales Database

P: High

Cost: 4 Days



Basic Camera Feed Display

P: High

Cost: 2 Days



Basic GUI Setup

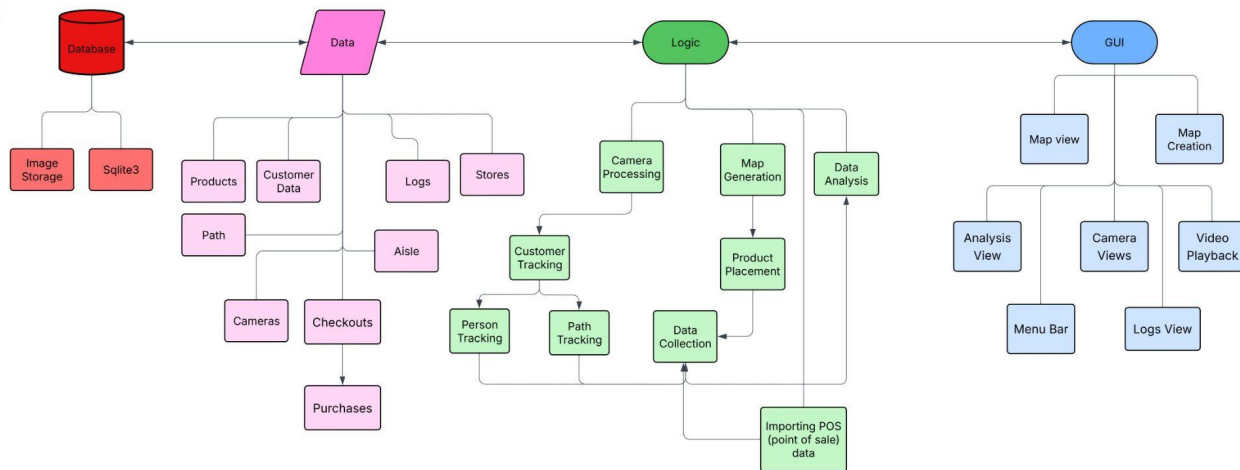
P: High

Cost: 3 Days

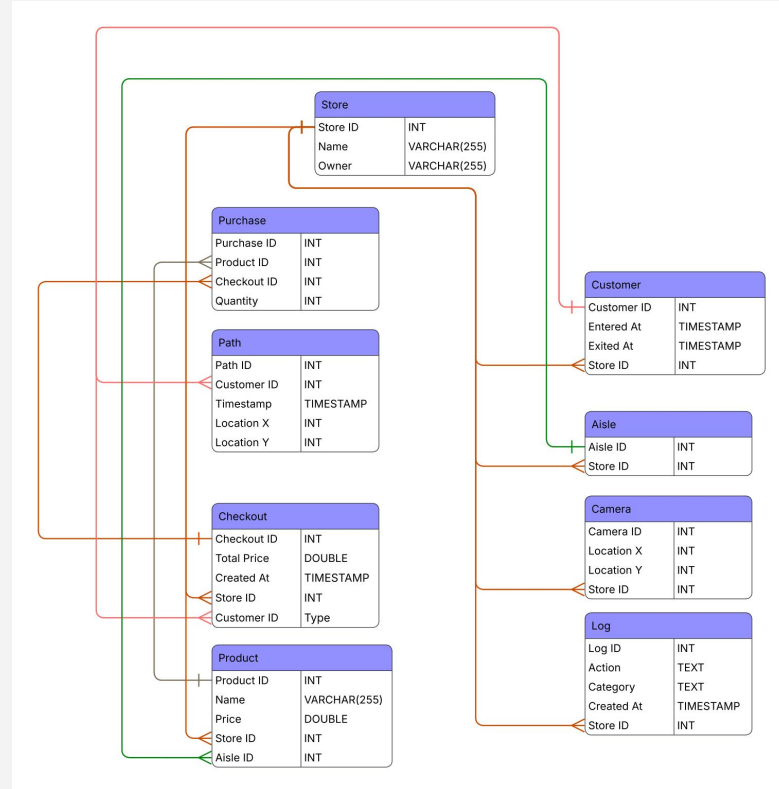




Development/Demo – Architecture



Development/Demo – Database (ITR0-ITR1)





Development/Demo – User Feedback

“The camera thing is pretty cool, I still understand it is very much a WIP. Look forward to seeing it develop.” – James Longerfield





Conclusion

Done Correctly

- Make time for each other
- Work review and quality assurance

Things to Improve

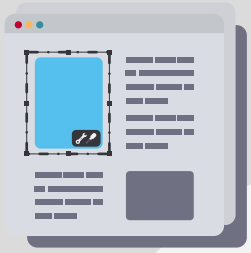
- Keep more detailed logs, use AI to create concise and rapid summaries

Plan for ITR2 – Data Tracking

- All about data gathering and putting it in the database
- Changed scope and features
 - Multiple camera angles → Top-down angle and stitch camera feeds
 - No more live feeds, all processed after the fact
 - Added customer demographic analysis via pictures at checkout (never save pictures, just the data)
- Record path data of users

Plan for ITR3 – Data Processing

- All about data mining and graphically represent the data
- Market Basket analysis
- Slow-zone displays
- Demographic analysis



Thanks!

